```javascript
function search(nums, target) {
    let left = 0;
    let right = nums.length - 1;

    while (left <= right) {
        let mid = Math.floor((left + right) / 2);

        if (nums[mid] === target) {
            return mid;
        }

        // Left half is sorted
        if (nums[left] <= nums[mid]) {
            if (target >= nums[left] && target < nums[mid]) {
                right = mid - 1;
            } else {
                left = mid + 1;
            }
        }
        // Right half is sorted
        else {
            if (target > nums[mid] && target <= nums[right]) {
                left = mid + 1;
            } else {
                right = mid - 1;
            }
        }
    }

    return -1;
}

console.log(search([4,5,6,7,0,1,2], 0));
```


```javascript
function search(nums, target) {
    let left = 0;
    let right = nums.length - 1;

    while (left <= right) {
        let mid = Math.floor((left + right) / 2);

        if (nums[mid] === target) {
            return mid;
        }

        // Left half is sorted
        if (nums[left] <= nums[mid]) {
            if (target >= nums[left] && target < nums[mid]) {
                right = mid - 1;
            } else {
                left = mid + 1;
            }
        }
        // Right half is sorted
```


```
PS C:\Users\Rindhiya\OneDrive\Desktop\need> node script.js
4
PS C:\Users\Rindhiya\OneDrive\Desktop\need>
```

```javascript
var maxArea = function(height) {
    let left = 0;
    let right = height.length - 1;
    let maxWater = 0;

    while (left < right) {
        let h = Math.min(height[left], height[right]);
        let width = right - left;
        let area = h * width;

        maxWater = Math.max(maxWater, area);

        // Move smaller height pointer
        if (height[left] < height[right]) {
            left++;
        } else {
            right--;
        }
    }

    return maxWater;
};
console.log(maxArea([1,8,6,2,5,4,8,3,7])); // 49
console.log(maxArea([1,1])); // 1
```

```javascript
var maxArea = function(height) {
    let left = 0;
    let right = height.length - 1;
    let maxWater = 0;

    while (left < right) {
        let h = Math.min(height[left], height[right]);
        let width = right - left;
        let area = h * width;

        maxWater = Math.max(maxWater, area);

        // Move smaller height pointer
        if (height[left] < height[right]) {
            left++;
        } else {
            right--;
        }
    }
```

```
PS C:\Users\Rindhiya\OneDrive\Desktop\need> node script.js
49
1
PS C:\Users\Rindhiya\OneDrive\Desktop\need>
```

```javascript
var canCompleteCircuit = function(gas, cost) {
    let totalTank = 0;
    let currentTank = 0;
    let start = 0;

    for (let i = 0; i < gas.length; i++) {
        let diff = gas[i] - cost[i];

        totalTank += diff;
        currentTank += diff;

        // If we can't continue, reset start
        if (currentTank < 0) {
            start = i + 1;
            currentTank = 0;
        }
    }

    return totalTank >= 0 ? start : -1;
};
console.log(canCompleteCircuit([1,2,3,4,5], [3,4,5,1,2])); // 3
console.log(canCompleteCircuit([2,3,4], [3,4,3])); // -1
```

```javascript
var canCompleteCircuit = function(gas, cost) {
    let totalTank = 0;
    let currentTank = 0;
    let start = 0;

    for (let i = 0; i < gas.length; i++) {
        let diff = gas[i] - cost[i];

        totalTank += diff;
        currentTank += diff;

        // If we can't continue, reset start
        if (currentTank < 0) {
            start = i + 1;
            currentTank = 0;
        }
    }

    return totalTank >= 0 ? start : -1;
};
```

```
PS C:\Users\Rindhiya\OneDrive\Desktop\need> node script.js
3
-1
PS C:\Users\Rindhiya\OneDrive\Desktop\need>
```